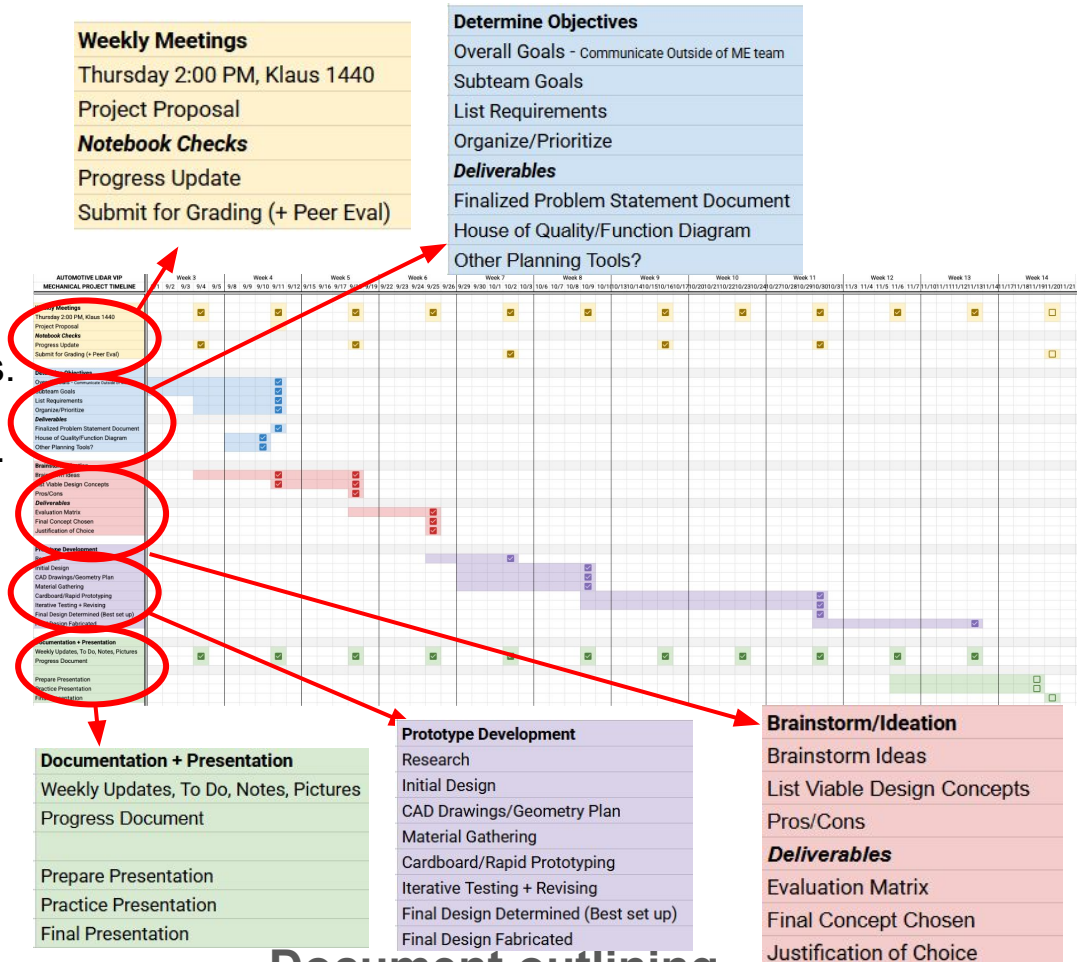


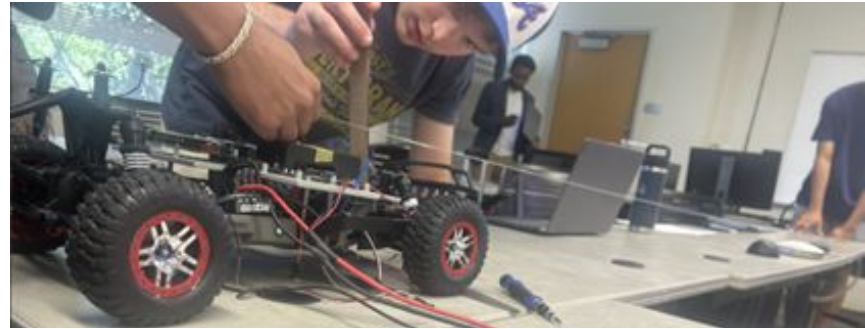
Overall Goal/Timeline

- Manufacture a LiDAR Mechanism that would allow:
 - Additional rotation around the y axis.
 - Visibility of ten feet from the vehicle.
 - Forward and Reverse Navigation.
- Timeline
 - Establish requirements- 9/11
 - Final Concept Chosen- 9/25
 - Initial Design/CAD Complete- 10/09
 - Final Design Fabrication- 11/13



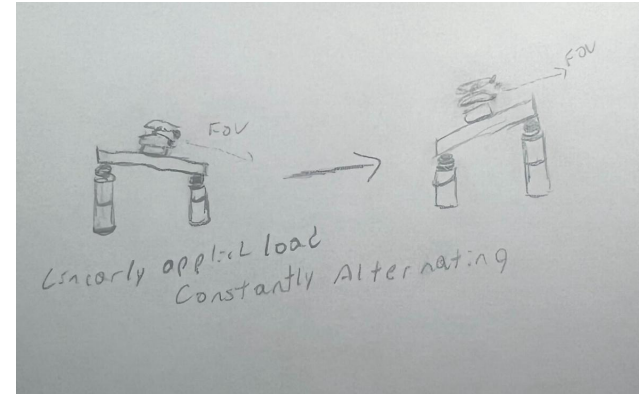
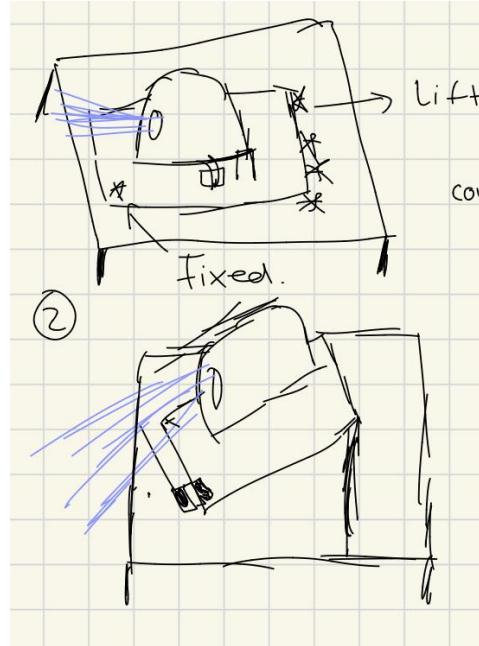
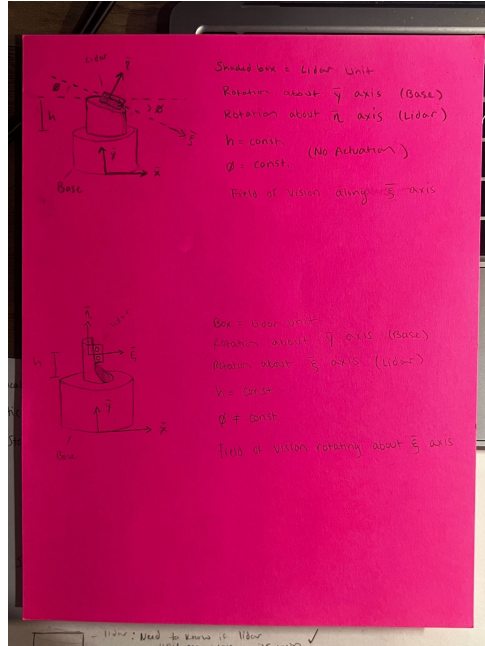
Requirements/Consideration

- LiDAR Height/Angle of Elevation
 - Minimum forward visibility of ten feet
 - Minimizes center of mass height
- Placement on Vehicle
 - Center of Mass is close to center of vehicle
 - Positions allows for forward and reverse navigation.
- Motor's for Rotation
- How to fasten the lidar/motors



**Measuring Field of View to find
Minimum LiDAR Height**

Initial Brainstorm/Drawings



Three different rough sketches of LiDAR mechanism

Chosen Concept

- Flip LiDAR and Motor 90 degrees
- Two different motors
 - One horizontal motor moves LiDAR left to right
 - One vertical motor moves LiDAR up & down

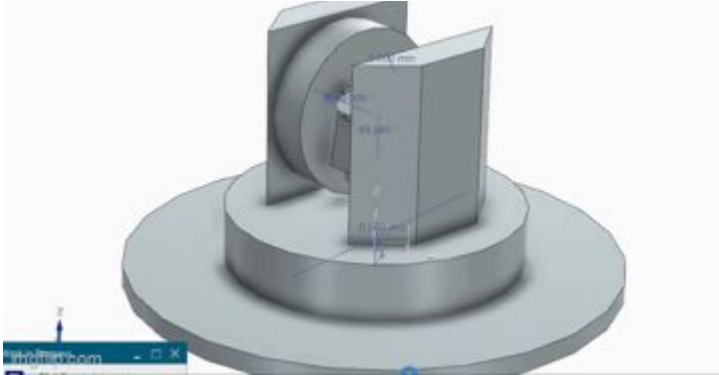
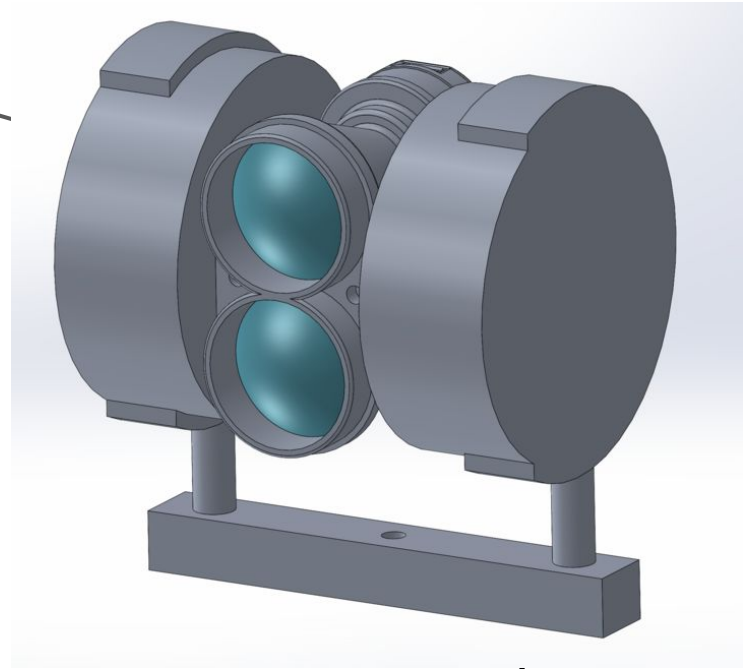


Image Showing Motion of LiDAR

Vertical Motor



LiDAR

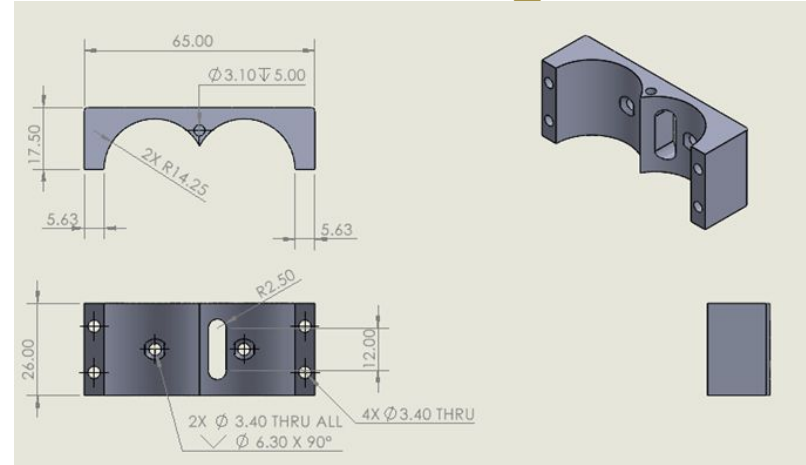
Initial LiDAR CAD of basic concept.

Initial Prototypes

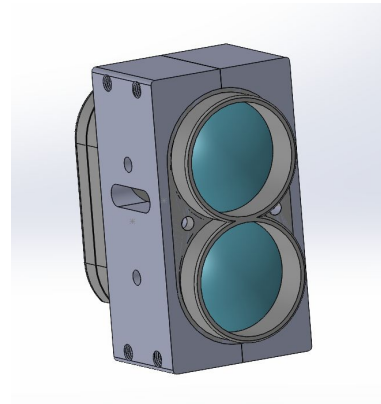


Initial Base and Motor Mounts

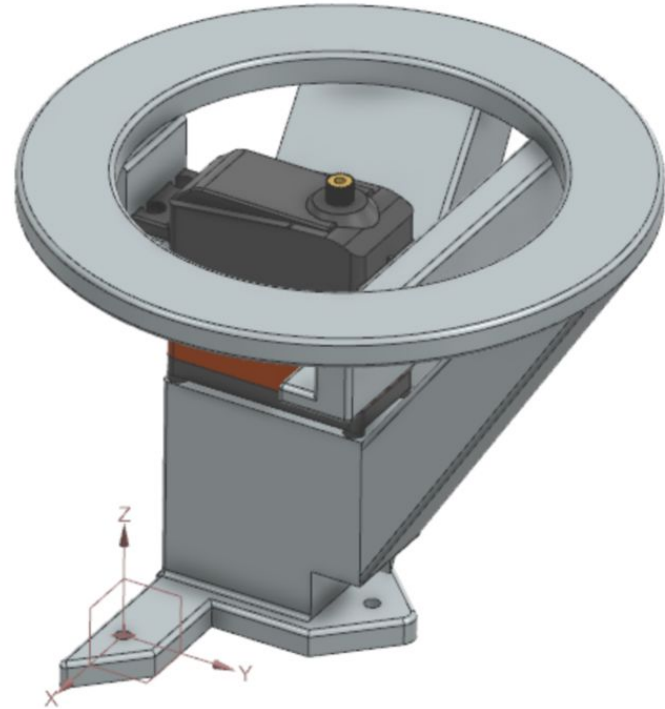
Final CAD Designs



LiDAR Clamp



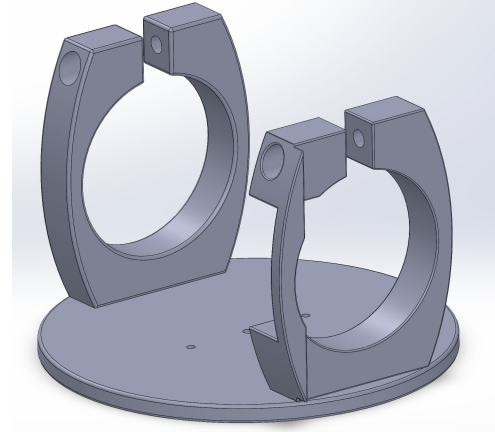
Servo Motor Mount



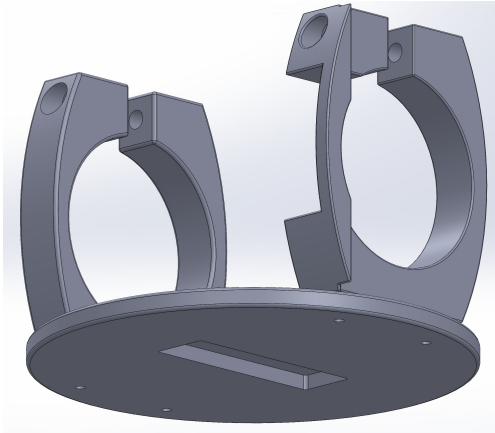
Final CAD Design



Gimbal Motor Clamp

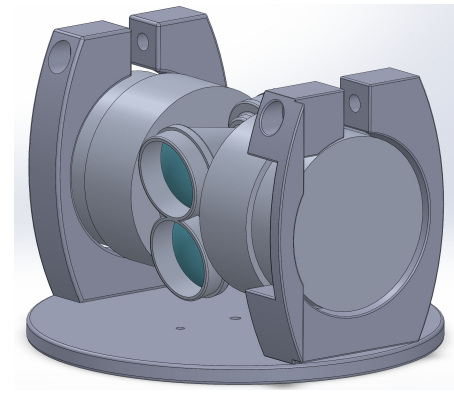
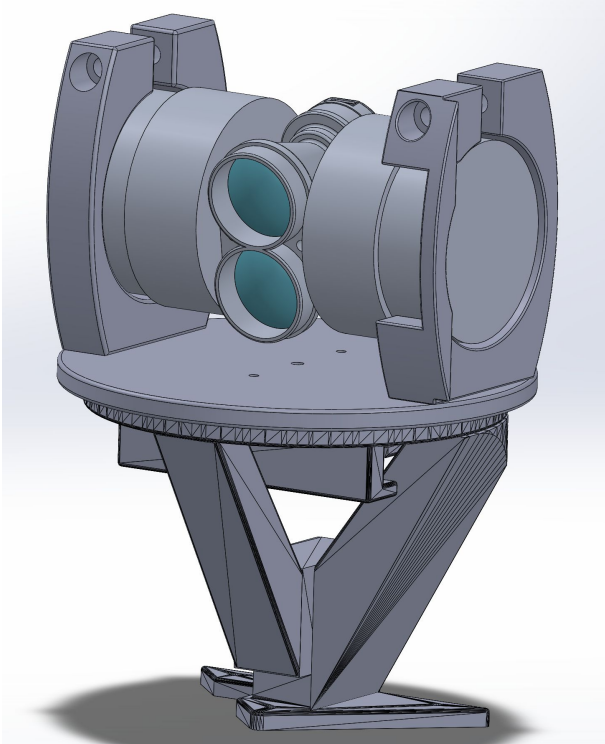


Gimbal Motor Clamp and
LiDAR Base Assembly

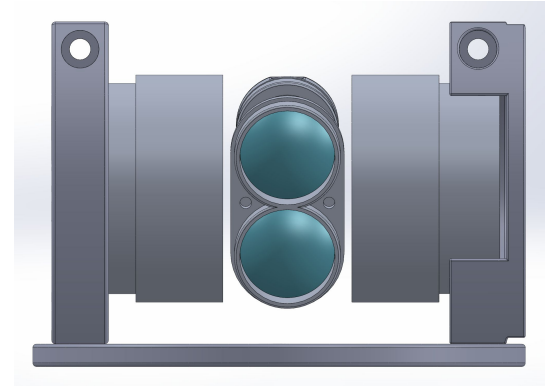


Final Assembled Design.

All 3 components assembled together

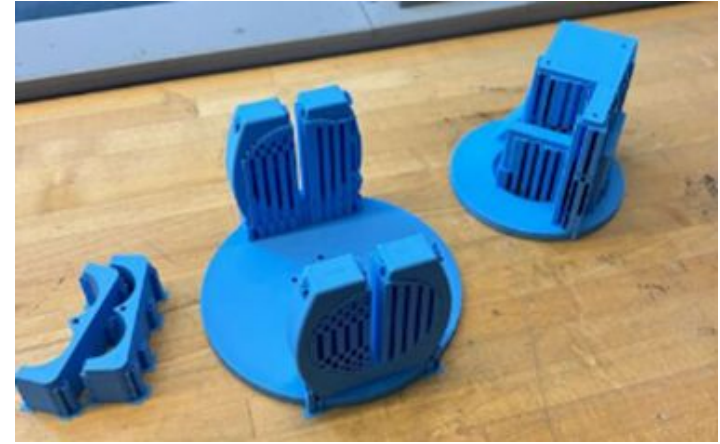


Gimbal Clamp, Base,
LiDAR, and Motor Assembly



Challenges Faced

- Clamping Gimbal Motor
 - Solution: Use split clamp design
- Adjustments with sizing
 - Solution: Repeated CAD and 3D printing
- Available space/mounting points on the base plate
 - Solution: CAD to create accurate dimensions and placement.
- Print Quality of Parts
 - Solution: Minimize overhang and orient parts to minimize supports



**Initial 3D Prints of Parts
with Supports**

Goals for Next Semester

- Getting a view of the back of the car
- Optimizing our design if possible
- Connect the spinning base to the motor

